WORKED SOLUTION

STEP I 1997, Question 10

Mechanics • Circular Motion

STEP I 1997, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s1-q10

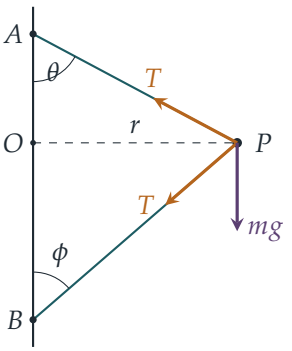
The key idea

Because the ring is smooth, the string runs freely through it, so the tension is the *same* on both sides of P. Resolve vertically (no vertical acceleration) and horizontally (acceleration $r\omega^2$ towards the axis), then divide one equation by the other so that T and m cancel. The sum-to-product formulae then turn the ratio into a single tangent of a half-angle.

Solution

The model and the forces

We treat P as a particle on a light inextensible string; the ring is smooth, so the tension T is the same in PA and PB. Let O be the centre of the circle, on the line AB with $OP = r$. The forces on P are its weight mg, a tension T along PA and a tension T along PB.



The string PA makes angle θ with the vertical (the angle at A between AB and AP), so its pull on P has components $T\cos\theta$ upwards and $T\sin\theta$ towards the axis. Similarly PB makes angle ϕ with the vertical at B, and its pull on P has components $T\cos\phi$

downwards and $T\sin\phi$ towards the axis.

The equations of motion

P moves in a horizontal circle at constant speed, so it has no vertical acceleration and an acceleration $r\omega^2$ towards O. Resolving,

$$\begin{aligned} \text{vertically:} \quad & T\cos\theta - T\cos\phi = mg, \\ \text{horizontally:} \quad & T\sin\theta + T\sin\phi = mr\omega^2. \end{aligned}$$

These hold whatever the shape of the triangle (if P were above A, say, $\cos\theta$ would simply be negative). Since $mg > 0$, the first equation gives $\cos\theta > \cos\phi$, so $\phi > \theta$: the lower string is the more steeply inclined to the axis, as the diagram suggests.

Showing the result

Dividing the vertical equation by the horizontal one,

$$\frac{\cos\theta - \cos\phi}{\sin\theta + \sin\phi} = \frac{g}{r\omega^2}.$$

By the sum-to-product formulae,

$$\begin{aligned} \cos\theta - \cos\phi &= 2\sin\tfrac{1}{2}(\theta+\phi)\,\sin\tfrac{1}{2}(\phi-\theta), \\ \sin\theta + \sin\phi &= 2\sin\tfrac{1}{2}(\theta+\phi)\,\cos\tfrac{1}{2}(\phi-\theta). \end{aligned}$$

Since $0 < \theta < \pi$ and $0 < \phi < \pi$ (they are angles of a genuine triangle APB), we have $0 < \frac{1}{2}(\theta+\phi) < \pi$, so $\sin\frac{1}{2}(\theta+\phi) \neq 0$ and it cancels:

$$\tan\frac{\phi-\theta}{2} = \frac{g}{r\omega^2},$$

as required.

The tension

Ask yourself The answer may contain $\sin\frac{1}{2}(\theta+\phi)$ but not $\phi-\theta$. How do we get rid of $\cos\frac{1}{2}(\phi-\theta)$?

From the horizontal equation and the second sum-to-product formula,

$$2T\sin\tfrac{1}{2}(\theta+\phi)\cos\tfrac{1}{2}(\phi-\theta) = mr\omega^2.$$

We know the *tangent* of $\alpha = \frac{1}{2}(\phi-\theta)$, so we can find its cosine from a right-angled triangle with opposite side g and adjacent side $r\omega^2$. We have $0 < \phi-\theta < \pi$, so $0 < \alpha < \frac{\pi}{2}$ and $\cos\alpha > 0$:

$$\cos\tfrac{1}{2}(\phi-\theta) = \frac{r\omega^2}{\sqrt{r^2\omega^4 + g^2}}.$$

Substituting and cancelling $r\omega^2$,

$$T = \frac{m\sqrt{r^2\omega^4 + g^2}}{2\sin\frac{1}{2}(\theta+\phi)}.$$

There is a nice way to see this: $\sqrt{(mg)^2 + (mr\omega^2)^2}$ is the size of the resultant of the two tensions, and two equal forces T at angle $\pi - (\theta+\phi)$ to each other (the angle APB) have a resultant of size $2T\sin\frac{1}{2}(\theta+\phi)$.

Checks

- As $\omega \to \infty$ the right-hand side tends to 0, so $\phi \to \theta$. The ring is flung out until it sits level with the midpoint of AB, as we'd expect when gravity is negligible.
- As $\omega \to 0$, $\tan\frac{1}{2}(\phi-\theta) \to \infty$, so $\phi - \theta \to \pi$: the string hangs straight down, which is the only possibility with no rotation.
- Numerically, we placed A, B and P at random, found T and ω^2 directly from the vector equation of motion, measured θ and ϕ from the coordinates, and compared with both results: the worst relative error over about 8000 cases was 3×10^{-12}.

Common slips

- Using two different tensions T_1, T_2. With a smooth ring there is one string and one tension; with two different tensions the problem is not determined.
- Getting the sign of the vertical component of the tension in PB wrong: that string pulls P *down* towards B.
- Taking the negative square root for $\cos\frac{1}{2}(\phi-\theta)$, which would give a negative tension.

Beyond STEP

For a given string length, the ring lies on an ellipse with foci A and B, and the string makes equal angles with the tangent at P: the reflection property of the ellipse. So the resultant of the tensions is along the normal to the ellipse, just like the reaction of a smooth wire bent into that shape. Thinking of constraints as forces normal to a surface is the starting point of Lagrangian mechanics. Kleppner and Kolenkow, *An Introduction to Mechanics*, has many problems on rotating systems of this kind.

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